Cabinet Installation & Scaling Guide

This guide describes how to deploy the Pumping Station IoT monitoring hardware inside electrical cabinets, activate cellular SIM cards, and scale the system to 400+ stations.

📶 Section 1: SIM Card Activation (1NCE M2M)

The pilot uses a **1NCE IoT 4G SIM Card** (recommended flat rate for €10/10 years).

1. **Register**: Go to the 1NCE Customer Portal and create an account.
2. **Activate SIM**: Click **Activate SIM Card** and enter the **ICCID** and **Activation Code** printed on the plastic SIM holder.
3. **Wait for network allocation**: Activation typically takes 10 to 30 minutes. Once activated, insert the SIM card into the **Micro-SIM slot** of the Waveshare ESP32-S3 board (ensure the golden contacts face downwards toward the PCB).

Note: The firmware configuration APN is already set to `iot.1nce.net` (no username/password needed).

🛠 Section 2: Cabinet Hardware Installation

Follow these guidelines for physical deployment inside the pump station electrical panel:

1. Enclosure Mounting

- Mount the Waveshare board inside a **waterproof IP65 plastic enclosure** with pre-drilled cable glands.
- Secure the enclosure inside the electrical cabinet using double-sided mounting tape or DIN-rail clips.
- *Caution:* Do not use a metal enclosure for the board, as it will block 4G cellular signals.

2. Antenna Placement

- Screw the external **4G cellular antenna** onto the board's gold SMA connector.
- Route the antenna cable out of the enclosure. If the main electrical cabinet is metal, route the antenna *outside* the metal cabinet (using a rubber-gasket hole) or place the antenna close to non-metallic windows or vents.
- In weak signal areas, mount a high-gain whip antenna externally.

3. Power Supply Connection

- Provide a stable **5V DC** supply to the ESP32 board.
- The safest method is placing a **5V / 2A DIN-rail power supply** (e.g., Mean Well MDR-10-5) inside the cabinet, wiring it to the 230V AC lines, and connecting its 5V output to the board's 5V/GND headers or USB-C port.

4. CT Clamp Installation

- Clip the SCT-013-020 current clamp around the **pump's active Phase cable** (L1 or L2 or L3) inside the cabinet.

- Connect the clamp's jack into your bias circuit box, then route the signal to GPIO 1.

🚀 Section 3: Flashing and Scaling Steps (For 400+ Stations)

Once the template is established, you can clone and provision new stations yourself in less than 5 minutes per device.

Workflow Diagram

```
[Admin Portal Dashboard] ──(Create new station)──▶ [Generates Provision Token]
                                                                  │
[Physical ESP32 Board] ◀──(Flash firmware with token)─────────────┘
```

Step-by-Step Cloning Procedure

1. **Provision on the Web Dashboard**:
   - Log in to the Cloud Dashboard as an Administrator.
   - Navigate to **Stationen verwalten** (Station Management) and click **Station hinzufügen** (Add Station).
   - Enter a unique ID (e.g., `STATION_002`), name, GPS coordinates, and pump power.
   - Click **Provisionieren**. The screen will display a long character sequence called the **Provisionierungstoken** (Provisioning Token). Copy this token.

2. **Configure the Firmware**:
   - Open the firmware code folder in PlatformIO (VS Code).
   - Open `src/config.h`.
   - Update the following lines with the values you just created:

     ```
     #define DEFAULT_STATION_ID    "STATION_002" // The new ID you added
     #define DEFAULT_CUSTOM_TOKEN  "PASTE_THE_COPIED_TOKEN_HERE"
     ```

   - *Ensure the APN is still set to `iot.1nce.net` (for 1NCE SIMs).*

3. **Flash the Board**:
   - Connect the new Waveshare ESP32-S3 board to your computer using a USB-C cable.
   - In PlatformIO, click the **Upload** arrow at the bottom status bar.
   - Once compilation finishes and the terminal outputs `SUCCESS`, unplug the board.

4. **Deploy & Verify**:
   - Power up the board. The status LED will cycle:
     - **Solid ON**: Booting.
     - **Slow Blink**: Connecting to cellular 4G network.

    - **Fast Blink**: Contacting Firebase authentication and exchanging the provisioning token for a permanent key.
    - **Heartbeat (double pulse)**: Normal operation.
  - Go to the Dashboard map or list view. The new station will show as 🟢 **Normal** (Online) and start streaming live current readings.
  - Edit the High/Low current thresholds for this station directly on the dashboard tab.